



3324 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	GC_MIRI_full	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	2	GC_background_MIRI	MIRI Medium Resolution Spectroscopy	(5) GC-MIRI-BKGND
	3	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	4	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	5	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	6	GC_MIRI_full	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	7	GC_background_MIRI	MIRI Medium Resolution Spectroscopy	(5) GC-MIRI-BKGND
	8	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	9	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	10	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	11	GC_MIRI_full	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	12	GC_background_MIRI	MIRI Medium Resolution Spectroscopy	(5) GC-MIRI-BKGND
	13	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	14	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A
	15	Staring mode 1	MIRI Medium Resolution Spectroscopy	(1) NAME-SGR-A

ABSTRACT

The Galactic Center offers the closest opportunity for studying accretion onto a supermassive black hole, Sagittarius A*. Sgr A*'s fluctuating emission is detected across the electromagnetic spectrum and may originate in the accretion flow or jet. Recent models indicate that variability can be produced by particle acceleration in different types of reconnection events. These models produce unique spectral characteristics imprinted on the variable emission in the MIR. Measurements of the spectral energy distribution in this wavelength regime during quiescent and flaring episodes in concert with X-ray monitoring will enable us to distinguish between models and constrain the energy budget of the emission region. In addition, the circumnuclear environment of Sgr A* contains dozens of IR sources and structures that are critical to understanding both the accretion-feeding processes and the nature of the cluster of stars and dusty gas features near it. We propose MIRI MRS IFU time-series spectral monitoring of Sgr A* and its surrounding region, along with simultaneous Chandra observations, to characterize the black hole's MIR and X-ray emission in different flux density states, determine the nature of the emission, and constrain models of the accretion and radiation processes. Only JWST has the high angular resolution and sensitivity in the mid-IR to probe this complex and dynamic region.

OBSERVING DESCRIPTION

The goal of this program is to obtain time-series MIRI MRS IFU spectra of the region centered on Sgr A* in order to determine the spectra of the time-variable emission from the region near the supermassive black hole.

We will obtain a full MIRI spectrum from 5 to 28 microns of the field including Sgr A* with MIRI MRS at the beginning of the monitoring period using the conventional observing strategy with four dithers in each of three grating settings to cover the full spectral range. Observations will use the fast readout mode to avoid saturation. A single exposure in each of four dither positions will give good signal to noise. A second nearby position will be observed as a background reference field.

The majority of the time will be spent in staring mode at the shortest MIRI MRS grating setting to monitor Sgr A* and detect fluctuations in the signal. The time is divided into three epochs, each containing three staring observations that include about 7 hours of science time, simultaneous with Chandra observations. Based on previous ground-based and Spitzer observations, we expect that during the monitoring period there will be one peak or flare with an emission level of ~ 5 mJy at 5 microns, several peaks at the 2 mJy level, and many smaller fluctuations that we will be able to detect, at least at the shorter wavelengths, at higher S/N than previous observations and in a new spectral regime.

Proposal 3324 - Targets - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000		
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes] Extended=NO				
Fixed Targets	(5)	GC-MIRI-BKGND	RA: 17 45 42.4279 (266.4267829d) Dec: -29 00 36.72 (-29.01020d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 3324 - Observation 1 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 1: GC_MIRI_full													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Star													
	Description=[Black holes]													
Extended=NO														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
	FND	ALL				NO				FULL				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	2	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	2	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	3	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	3	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	

Proposal 3324 - Observation 1 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Special Requirements	Between Dates 10-MAR-2024:00:00:00 and 05-MAY-2024:00:00:00
	Sequence Observations 1, 2, 3, 4, 5, Non-interruptible

Proposal 3324 - Observation 2 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 2: GC_background_MIRI													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(5)	GC-MIRI-BKGND	RA: 17 45 42.4279 (266.4267829d) Dec: -29 00 36.72 (-29.01020d) Equinox: J2000					Epoch of Position: 2000.0						
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel					Simultaneous Imaging				Imager Subarray			
	FND	ALL					NO				FULL			
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	

Proposal 3324 - Observation 2 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Special Requirements	Sequence Observations 1, 2, 3, 4, 5, Non-interruptible
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Proposal 3324 - Observation 3 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 3: Staring mode 1													Thu May 11 00:10:30 GMT 2023		
	Diagnostic Status: Warning															
	Observing Template: MIRI Medium Resolution Spectroscopy															
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold															
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.															
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous				
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000													
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.															
	Category=Star															
	Description=[Black holes]															
Acquisition	Extended=NO															
	#														Target	
	1														NONE	
Template	AcqFilter		Primary Channel					Simultaneous Imaging				Imager Subarray				
	F560W		ALL					NO				FULL				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144			
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144			
Special Requirements	Time Series Observation															
	No Parallel Attachments															
	No Parallel Attachments															
	Sequence Observations 1, 2, 3, 4, 5, Non-interruptible															

Proposal 3324 - Observation 4 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 4: Staring mode 1													Thu May 11 00:10:30 GMT 2023		
	Diagnostic Status: Warning															
	Observing Template: MIRI Medium Resolution Spectroscopy															
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over lower threshold															
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.															
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous					
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000													
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.															
	Category=Star															
	Description=[Black holes]															
Acquisition	Extended=NO															
	#														Target	
	1														NONE	
Template	AcqFilter		Primary Channel				Simultaneous Imaging				Imager Subarray					
	F560W		ALL				NO				FULL					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144			
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144			
Special Requirements	Time Series Observation															
	No Parallel Attachments															
	No Parallel Attachments															
	Sequence Observations 1, 2, 3, 4, 5, Non-interruptible															

Proposal 3324 - Observation 5 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 5: Staring mode 1												Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Data Excess over lower threshold												
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Star												
	Description=[Black holes]												
Acquisition	Extended=NO												
	#	Target											
	1	NONE											
Template	AcqFilter		Primary Channel				Simultaneous Imaging			Imager Subarray			
	F560W		ALL				NO			FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments												
	Sequence Observations 1, 2, 3, 4, 5, Non-interruptible												

Proposal 3324 - Observation 6 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 6: GC_MIRI_full													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Visit 6:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Star													
	Description=[Black holes]													
Extended=NO														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
	FND	ALL				NO				FULL				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Proposal 3324 - Observation 6 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Special Requirements	Between Dates 10-MAR-2024:00:00:00 and 05-MAY-2024:00:00:00
	Sequence Observations 6, 7, 8, 9, 10, Non-interruptible

Proposal 3324 - Observation 7 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 7: GC_background_MIRI													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(5)	GC-MIRI-BKGND	RA: 17 45 42.4279 (266.4267829d) Dec: -29 00 36.72 (-29.01020d) Equinox: J2000					Epoch of Position: 2000.0						
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel					Simultaneous Imaging				Imager Subarray			
	FND	ALL					NO				FULL			
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Proposal 3324 - Observation 7 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Special Requirements	Sequence Observations 6, 7, 8, 9, 10, Non-interruptible
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Proposal 3324 - Observation 8 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 8: Staring mode 1													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 8:1) Warning (Form): Data Excess over lower threshold													
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Star													
	Description=[Black holes]													
Acquisition	Extended=NO													
	#	Target												
	1	NONE												
Template	AcqFilter		Primary Channel					Simultaneous Imaging			Imager Subarray			
	F560W		ALL					NO			FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144	
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144	
Special Requirements	Time Series Observation													
	No Parallel Attachments													
	No Parallel Attachments													
	Sequence Observations 6, 7, 8, 9, 10, Non-interruptible													

Proposal 3324 - Observation 9 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 9: Staring mode 1													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 9:1) Warning (Form): Data Excess over lower threshold													
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Star													
	Description=[Black holes]													
Acquisition	Extended=NO													
	#	Target												
	1	NONE												
Template	AcqFilter		Primary Channel					Simultaneous Imaging			Imager Subarray			
	F560W		ALL					NO			FULL			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144	
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144	
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments													
	Sequence Observations 6, 7, 8, 9, 10, Non-interruptible													

Proposal 3324 - Observation 10 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 10: Staring mode 1													Thu May 11 00:10:30 GMT 2023	
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 10:1) Warning (Form): Data Excess over lower threshold														
	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous				
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d) Dec: -29 00 28.17 (-29.00783d) Equinox: J2000												
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.														
	Category=Star Description=[Black holes] Extended=NO														
Acquisition	#														Target
	1														NONE
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray					
	F560W	ALL				NO				FULL					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbkbk.Calc ID		
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144		
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144		
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments														
	Sequence Observations 6, 7, 8, 9, 10, Non-interruptible														

Proposal 3324 - Observation 11 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 11: GC_MIRI_full													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Visit 11:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d)											
			Dec: -29 00 28.17 (-29.00783d)											
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Acquisition	Category=Star													
	Description=[Black holes]													
	Extended=NO													
Template	#						Target							
	1						NONE							
Dithers	AcqFilter		Primary Channel					Simultaneous Imaging			Imager Subarray			
	FND		ALL					NO			FULL			
Spectral Elements	#	Dither Type					Optimized For			Direction				
	1	4-Point					POINT SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Proposal 3324 - Observation 11 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?	
Special Requirements	Between Dates 10-MAR-2024:00:00:00 and 05-MAY-2024:00:00:00
	Sequence Observations 11, 12, 13, 14, 15, Non-interruptible

Proposal 3324 - Observation 12 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 12: GC_background_MIRI													Thu May 11 00:10:30 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	GC-MIRI-BKGND	RA: 17 45 42.4279 (266.4267829d) Dec: -29 00 36.72 (-29.01020d) Equinox: J2000				Epoch of Position: 2000.0							
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
	FND	ALL				NO				FULL				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	146144	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Proposal 3324 - Observation 12 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Special Requirements	Sequence Observations 11, 12, 13, 14, 15, Non-interruptible
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Proposal 3324 - Observation 13 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 13: Staring mode 1													Thu May 11 00:10:30 GMT 2023	
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 13:1) Warning (Form): Data Excess over lower threshold														
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous				
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d)												
			Dec: -29 00 28.17 (-29.00783d)												
			Equinox: J2000												
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes] Extended=NO													
Acquisition	#														Target
	1														NONE
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray					
	F560W	ALL				NO				FULL					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID		
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144		
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144		
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments														
	Sequence Observations 11, 12, 13, 14, 15, Non-interruptible														

Proposal 3324 - Observation 14 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 14: Staring mode 1													Thu May 11 00:10:30 GMT 2023		
	Diagnostic Status: Warning															
	Observing Template: MIRI Medium Resolution Spectroscopy															
Diagnostics	(Visit 14:1) Warning (Form): Data Excess over lower threshold															
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.															
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous				
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d)													
			Dec: -29 00 28.17 (-29.00783d)													
			Equinox: J2000													
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes] Extended=NO														
Acquisition	#														Target	
	1														NONE	
Template	AcqFilter		Primary Channel					Simultaneous Imaging				Imager Subarray				
	F560W		ALL					NO				FULL				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID			
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144			
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144			
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments															
	Sequence Observations 11, 12, 13, 14, 15, Non-interruptible															

Proposal 3324 - Observation 15 - Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?

Observation	Proposal 3324, Observation 15: Staring mode 1													Thu May 11 00:10:30 GMT 2023	
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 15:1) Warning (Form): Data Excess over lower threshold														
	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous			
	(1)	NAME-SGR-A	RA: 17 45 40.0360 (266.4168167d)												
			Dec: -29 00 28.17 (-29.00783d)												
			Equinox: J2000												
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Black holes] Extended=NO													
Acquisition	#											Target			
	1											NONE			
Template	AcqFilter	Primary Channel					Simultaneous Imaging				Imager Subarray				
	F560W	ALL					NO				FULL				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	SHORT(A)	MRSLONG		FASTR1	30	85	1	None	1	85	7309.455	146144		
	1	SHORT(A)	MRSSHORT		FASTR1	30	85	1	None	1	85	7309.455	146144		
Special Requirements	Time Series Observation No Parallel Attachments No Parallel Attachments														
	Sequence Observations 11, 12, 13, 14, 15, Non-interruptible														